

V-CURRENTS: Virtual Collaborative Underwater Realities for Engagement and Novel Teaching Spaces

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Figure 1: Underwater Virtual Field Trip, Snapshot of 360 video at Lake Grevelingen, The Netherlands. Source: Image, courtesy of Triton Oceanic Exploration Society; cartoon octopus, modified IEEE VR 2025 logo.

ABSTRACT

Ocean literacy—the understanding of how oceans influence the Earth’s ecosystems, their critical role in sustaining life on Earth, and the impact of human activities on the ocean—is essential for making informed decisions for addressing fundamental environmental problems. However, the inaccessibility of oceans brings about special challenges for educators and decision makers. We describe our work on V-CURRENTS, a platform for creating underwater virtual reality field trips as an approach to address some of these challenges and promote ocean literacy through immersive technologies and immersive experiences. We sketch key components of the platform as well as exemplary application scenarios.

Index Terms: Ocean literacy, underwater virtual field trips, medial transformations, virtual collaboration spaces.

1 INTRODUCTION

Ocean literacy is essential for fostering a deeper understanding of the ocean’s critical role in sustaining life on Earth [7]. It encompasses awareness of how the oceans influence the planet’s climate, weather, and ecosystems, as well as how human activities impact marine environments, for example, as part of the energy transition towards renewable sources. By promoting knowledge of ocean processes through experiences and demonstrating their interconnectedness with human health, food security, and economic prosper-

ity, ocean literacy empowers individuals and communities to make informed decisions that support the sustainable use and conservation of marine resources [4]. This is particularly crucial in the face of challenges such as climate change, overfishing, and pollution, which threaten the delicate balance of ocean ecosystems and have resulted in a dramatic loss of biodiversity. At the same time, developments such as the energy transition also offer opportunities for marine ecosystems to flourish and allow for creating new and protected spaces for marine fauna and flora. Ocean literacy can inspire stewardship and advocacy for marine environments by highlighting the importance of the ocean in cultural, economic, and scientific contexts. It equips educators, policymakers, and citizens with the tools to address global challenges, from reducing carbon emissions to protecting biodiversity. By fostering a sense of connection to the ocean, ocean literacy can encourage actions that benefit both marine ecosystems and future generations [2]. Ultimately, improving ocean literacy is not just about understanding the ocean; it is about fostering a sustainable relationship and stewardship between humanity and the planet’s most vital and expansive resource.

Immersive technologies, such as virtual reality, provide educators, policy-makers, stakeholders, and researchers with an opportunity to paradigmatically change how we learn and communicate about the oceans and advance ocean literacy [3]. Immersive technologies provide place-based education and decision-making even for locations or scenarios that are traditionally not accessible such as geological zones, the future, or the ocean floors [6]. Adding core features of virtual reality such as realistic experiences, interactivity, and collaboration allows for enhancing engagement and connecting people across locations and countries for a grounded approach to communication and decision-making [1]. These affordances of immersive experiences allow for facilitating ocean literacy in educational settings, for example, through virtual field trips, training (e.g., diving simulations), outreach, and citizen science activities.

In this contribution, we describe an emerging platform for designing collaborative underwater virtual field trips, which we call V-CURRENTS. This platform is part of a larger project on understanding the diverse aspects of the energy transition in the North

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Sea, focusing, for example, on emerging ecological opportunities.

2 V-CURRENTS

Over the years, many solutions for virtual field trips have been created on topics such as energy education, fire ecology, geoscience education, and others. At the same time, the science of virtual field trips and the pedagogy of immersive experiences have advanced [6]. We are building on these foundations and state-of-the-art technologies to deliver underwater virtual field trips and advance the pedagogy of virtual field trips to the oceans to give researchers, educators, stakeholders, and policy makers new perspectives. In abbreviated details, V-CURRENTS has the following components:

Collaborative, immersive experiences. Multi-user virtual trips raise a number of specific challenges that V-CURRENTS aims to address, including role management (e.g. instructor, student) as well as administration and communication tools to provide a smooth experience. One example are pointing/highlighting tools and guidance mechanism for both 3D and 360° media-based scenes as well as suitable for underwater scenarios with their specific characteristics (e.g., relevant objects tend to be further from away from the horizontal plane on average compared to traditional virtual field trips on land). The current implementation is based on RadishNet.

Content generation. One of our goals in V-CURRENT is to support a large variety of media to offer a flexible system for creating educational underwater experiences that, in addition, allows for efficient content creation by non-VR-experts. We are in the unique position to have access to 360° underwater videos recorded by one of the internationally leading underwater videographers, the Triton Oceanic Exploration Society (<https://www.tritonoes.com/>). Fig. 1 shows a screenshot of one of the videos that are currently part of V-CURRENTS, recorded in lake Grevelingen in the Netherlands. In addition, first experiences created with V-CURRENTS leverage developments from digital twinning approaches that will be adapted to the ocean [5] to complement the 360° videography.

Comprehensive assessment of user experiences. V-CURRENT is not only designed as a virtual tour generation tool but also as a research tool to evaluate the design and application of virtual field trips. The goal is to provide a comprehensive assessment environment that is recording user interactions, allows for administering integrated questionnaires, as well as support for the recording of behavioral data such as motion data and eye-tracking (built into headsets) and physiological signals such as galvanic skin resistance (though external sensors).

3 SCENARIOS

Examples of scenarios for our underwater virtual field trip include: (1) the different stages in the lifetime of an offshore wind farm in the North Sea: the pre-construction baseline with mostly sandy bottom and already some human activities (fishing, seismic exploration), the construction phase with the placement of the scour protection and piling of monopiles (with very loud sounds), and after that many species that use the new opportunities (hard substrate) that are provided through various stages of succession. First the colonization of biofilms and fast-growing pioneer species, followed by the establishment of filter feeders and macroalgae. Over time, competition leads to an increasingly diverse community, eventually maturing into a stable, biodiverse ecosystem with long-lived species. The last stage is an open - and soon very relevant question - what do we do with these structures after the lifetime of the wind farm (after 25 years) and at the same time being a thriving reef.

(2) As shown in the screenshot of the underwater 360° video in Fig. 1, recorded in lake Grevelingen (a saline lake and former arm of the North Sea), the outlet of a pumping station helps drain excess water from the polders around Dreischor into the lake during heavy rainfall. The structure around the outlet serves as a vertical substrate for reef-building Pacific oysters and blue mussels. Both

species are filter-feeders, consuming free-floating plankton and organic particles. By doing so, they improve water quality and clarity. These reef-building organisms create habitats that provide shelter and attachment surfaces for a variety of organisms, including tunicates (visible as orange blobs in the video (Fig. 1), sponges (white-yellow blobs), algae (red), crabs, and fish. The large shoal of free-swimming animals consists mostly of opossum shrimp, along with occasional small fish.

The location featured in the video is relatively shallow, allowing light to penetrate the water. In this photic zone, the benthic life in lake Grevelingen is relatively rich. However, deeper areas have suffered from the lake's separation from the North Sea. Without tidal exchange, these areas experience oxygen depletion, leaving behind mostly mud and bacteria.

4 OUTLOOK AND CONCLUSIONS

The United Nations have launched the decade of the oceans, drawing attention to their vital role in protecting and sustaining life on earth. The XR community has a unique opportunity to play a key role in advancing ocean literacy by creating place-based experiences for otherwise inaccessible locations and by using new levels of realism and interactivity to create engagement, emotional connections, and stewardship for the oceans. The V-CURRENTS platform is being designed to provide a multi-modal, and multi-media approach to integrate various sources and combine them with empirically validated immersive design principles.

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